

could simply have included `<QtGui>`. However, it is generally bad style to include such a big header file from another header file, especially in larger applications.

```

• 4      : QDialog(parent)
• 5 {
• 6     label = new QLabel(tr("Find &what:"));
• 7     lineEdit = new QLineEdit;
• 8     label->setBuddy(lineEdit);
• 9     caseCheckBox = new QCheckBox(tr("Match &case"));
• 10    backwardCheckBox = new QCheckBox(tr("Search &back-
ward"));
• 11    findButton = new QPushButton(tr("&Find"));
• 12    findButton->setDefault(true);
• 13    findButton->setEnabled(false);
• 14    closeButton = new QPushButton(tr("Close"));

```

On line 4, we pass on the parent parameter to the base class constructor. Then we create the child widgets. The `tr()` function calls around the string literals mark them for translation to other languages. The function is declared in `QObject` and every subclass that contains the `Q_OBJECT` macro. It's a good habit to surround user-visible strings with `tr()`, even if you don't have immediate plans for translating your applications to other languages.

In the string literals, we use ampersands ('&') to indicate shortcut keys. For example, line 11 creates a Find button, which the user can activate by pressing `Alt+F` on platforms that support shortcut keys. Ampersands can also be used to control focus: On line 6 we create a label with a shortcut key (`Alt+W`), and on line 8 we set the label's buddy to be the line editor. A buddy is a widget that accepts the focus when the label's shortcut key is pressed. So when the user presses `Alt+W` (the label's shortcut), the focus goes to the line editor (the label's buddy).

On line 12, we make the Find button the dialog's default button by calling `setDefault(true)`. The default button is the button that is pressed when the user hits `Enter`. On line 13, we disable the Find button. When a widget is disabled, it is usually shown grayed out and will not respond to user interaction.

```

• 15    connect(lineEdit, SIGNAL(textChanged(const QString
&)),
• 16        this, SLOT(enableFindButton(const QString
&)));

```